Lecture Notes: Confidence Intervals in Linear Regression, Pointwise Bands, and the Role of Leverage and Influence

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1 Overview

These notes develop the main confidence intervals in linear regression from first principles:

1. Confidence intervals for regression coefficients β_j
2. Pointwise confidence interval for the *mean response* $E[Y \mid X = x_0]$
3. Pointwise prediction interval for a *new observation* $Y_{\text{new}} \mid X = x_0$

We will show:

- why the prediction interval for $Y \mid X = x_0$ is wider than the confidence interval for $E[Y \mid X = x_0]$,
- why fitted values are more sensitive to points in the tails of the predictor space,
- how this sensitivity is encoded by *leverage*,
- how leverage combines with residual size to produce *Cook's distance*,
- how the width of pointwise intervals depends directly on leverage.

We include both proofs and numerical illustrations, along with R simulations.

2 Model Setup

Consider the classical linear regression model

$$\mathbf{Y} = X\beta + \varepsilon,$$

where:

- $\mathbf{Y}$ is an $n \times 1$ response vector,
- X is an $n \times p$ design matrix of full column rank,
- β is a $p \times 1$ unknown coefficient vector,
- $\varepsilon \sim N(0, \sigma^2 I_n)$.

The ordinary least squares estimator is

$$\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{Y}.$$

The fitted values are

$$\hat{\mathbf{Y}} = X\hat{\beta} = HY,$$

where

$$H = X(X^\top X)^{-1} X^\top$$

is the *hat matrix*.

3 Distribution of the OLS Estimator

3.1 Unbiasedness

Because

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y = (X^\top X)^{-1} X^\top (X\beta + \varepsilon) = \beta + (X^\top X)^{-1} X^\top \varepsilon,$$

we have

$$E[\hat{\beta}] = \beta + (X^\top X)^{-1} X^\top E[\varepsilon] = \beta.$$

Thus OLS is unbiased.

3.2 Variance

Using the above representation,

$$\mathrm{Var}(\hat{\beta}) = \mathrm{Var}\left((X^\top X)^{-1} X^\top \varepsilon\right).$$

Since $\mathrm{Var}(\varepsilon) = \sigma^2 I_n$,

$$\mathrm{Var}(\hat{\beta}) = (X^\top X)^{-1} X^\top (\sigma^2 I_n) X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

Therefore,

$$\hat{\beta} \sim N\left(\beta, \sigma^2 (X^\top X)^{-1}\right).$$

3.3 Componentwise form

For coefficient $\hat{\beta}_j$,

$$\mathrm{Var}(\hat{\beta}_j) = \sigma^2 c_{jj},$$

where c_{jj} is the jth diagonal entry of $(X^\top X)^{-1}$.

Hence

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 c_{jj}).$$

4 Confidence Interval for a Regression Coefficient

4.1 When σ^2 is known

If σ^2 were known, then

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

So a $100(1-\alpha)\%$ confidence interval is

$$\hat{\beta}_j \pm z_{\alpha/2}\sigma\sqrt{c_{jj}}.$$

4.2 When σ^2 is unknown

In practice,

$$\hat{\sigma}^2 = s^2 = \frac{\mathrm{SSE}}{n-p}, \qquad \mathrm{SSE} = (Y - X\hat{\beta})^\top (Y - X\hat{\beta}).$$

Under the normal linear model,

$$\frac{(n-p)s^2}{\sigma^2} \sim \chi^2_{n-p},$$

and this is independent of $\hat{\beta}$.

Therefore,

$$\frac{\hat{\beta}_j - \beta_j}{s\sqrt{c_{jj}}} \sim t_{n-p}.$$

Theorem 1 (CI for β_j)**.** *A $100(1-\alpha)\%$ confidence interval for β_j is*

$$\boxed{\hat{\beta}_j \pm t_{\alpha/2,n-p}\, s\sqrt{c_{jj}}}$$

4.3 Interpretation

This interval estimates the plausible values of the *population slope/intercept coefficient*. It is not an interval for a future observation.

5 Pointwise Confidence Interval for the Mean Response

Let x_0 be a $p \times 1$ vector of predictor values, including the intercept entry 1. The mean response at x_0 is

$$\mu(x_0) = E[Y \mid x_0] = x_0^\top \beta.$$

Its estimator is

$$\hat{\mu}(x_0) = x_0^\top \hat{\beta}.$$

5.1 Distribution of $\hat{\mu}(x_0)$

Since $\hat{\beta}$ is normal,

$$x_0^\top \hat{\beta} \sim N\left(x_0^\top \beta,\ \sigma^2 x_0^\top (X^\top X)^{-1} x_0\right).$$

Thus,

$$\mathrm{Var}(\hat{\mu}(x_0)) = \sigma^2 x_0^\top (X^\top X)^{-1} x_0.$$

Replacing σ by s gives

$$\frac{\hat{\mu}(x_0) - \mu(x_0)}{s\sqrt{x_0^\top (X^\top X)^{-1} x_0}} \sim t_{n-p}.$$

Theorem 2 (Pointwise CI for the mean response)**.** *A* $100(1-\alpha)\%$ *pointwise confidence interval for* $E[Y \mid x_0]$ *is*

$$\boxed{\hat{\mu}(x_0) \pm t_{\alpha/2,n-p}\, s\sqrt{x_0^\top (X^\top X)^{-1} x_0}}$$

5.2 In simple linear regression

In SLR with intercept,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

we take

$$x_0 = \begin{bmatrix} 1 \\ x_0 \end{bmatrix}.$$

Then

$$x_0^\top (X^\top X)^{-1} x_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}},$$

where

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Hence

$$\boxed{\hat{\mu}(x_0) \pm t_{\alpha/2,n-2}\, s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}}$$

6 Pointwise Prediction Interval for a New Observation

Now consider a *new* response at x_0:

$$Y_{\text{new}} = x_0^\top \beta + \varepsilon_{\text{new}}, \qquad \varepsilon_{\text{new}} \sim N(0, \sigma^2),$$

independent of the original sample.

We predict Y_{new} using $\hat{\mu}(x_0) = x_0^\top \hat{\beta}$.

6.1 Prediction error variance

The prediction error is

$$Y_{\text{new}} - \hat{\mu}(x_0) = x_0^\top \beta + \varepsilon_{\text{new}} - x_0^\top \hat{\beta}.$$

So

$$Y_{\text{new}} - \hat{\mu}(x_0) = \varepsilon_{\text{new}} - x_0^\top (\hat{\beta} - \beta).$$

Because ε_{new} is independent of $\hat{\beta}$,

$$\text{Var}(Y_{\text{new}} - \hat{\mu}(x_0)) = \text{Var}(\varepsilon_{\text{new}}) + \text{Var}(x_0^\top \hat{\beta}) = \sigma^2 + \sigma^2 x_0^\top (X^\top X)^{-1} x_0.$$

Therefore,

$$\text{Var}(Y_{\text{new}} - \hat{\mu}(x_0)) = \sigma^2 \left(1 + x_0^\top (X^\top X)^{-1} x_0\right).$$

So,

$$\frac{Y_{\text{new}} - \hat{\mu}(x_0)}{s\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}} \sim t_{n-p}.$$

Theorem 3 (Pointwise prediction interval)**.** *A $100(1-\alpha)\%$ pointwise prediction interval for a new response at x_0 is*

$$\boxed{\hat{\mu}(x_0) \pm t_{\alpha/2,n-p}\, s\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}}$$

7 Why the Prediction Interval Is Wider

Compare the two standard errors:

$$SE_{\text{mean}}(x_0) = s\sqrt{x_0^\top (X^\top X)^{-1} x_0},$$

$$SE_{\text{pred}}(x_0) = s\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}.$$

Since

$$1 + x_0^\top (X^\top X)^{-1} x_0 > x_0^\top (X^\top X)^{-1} x_0,$$

we have

$$SE_{\text{pred}}(x_0) > SE_{\text{mean}}(x_0).$$

Remark 1. *The mean-response interval reflects only uncertainty in estimating the regression line. The prediction interval must also include the irreducible noise of an individual response, represented by the extra "+1" term.*

7.1 Simple linear regression version

In SLR:

$$SE_{\text{mean}}(x_0) = s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}},$$

$$SE_{\text{pred}}(x_0) = s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

The only difference is the extra 1 inside the square root.

8 Numerical Example: Mean CI vs Prediction Interval

Suppose:

$$n = 20, \quad \bar{x} = 10, \quad S_{xx} = 200, \quad s = 4, \quad t_{0.025,18} \approx 2.10.$$

Take $x_0 = 10$ (the center of the data).

Then

$$\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} = \frac{1}{20} + 0 = 0.05.$$

So

$$SE_{\text{mean}}(10) = 4\sqrt{0.05} = 0.8944,$$

and the half-width is

$$2.10(0.8944) \approx 1.88.$$

For the prediction interval,

$$SE_{\text{pred}}(10) = 4\sqrt{1.05} \approx 4.099,$$

and the half-width is

$$2.10(4.099) \approx 8.61.$$

Thus the prediction interval is much wider.

8.1 At a tail value

Now take $x_0 = 20$.

Then

$$\frac{1}{n} + \frac{(20 - 10)^2}{200} = 0.05 + 0.50 = 0.55.$$

So

$$SE_{\text{mean}}(20) = 4\sqrt{0.55} \approx 2.966,$$

and the half-width is

$$2.10(2.966) \approx 6.23.$$

For a new observation,

$$SE_{\text{pred}}(20) = 4\sqrt{1.55} \approx 4.980,$$

and the half-width is

$$2.10(4.980) \approx 10.46.$$

8.2 Takeaway

Both intervals widen away from $\bar{x}$, but the prediction interval is always wider because it contains both:

- uncertainty in the fitted line,
- random noise of an individual outcome.

9 Why the Fitted Model Is More Sensitive Near the Tails

9.1 Sensitivity in simple linear regression

In SLR,

$$\hat{y}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

Since

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

we may write

$$\hat{y}(x_0) = \bar{y} + \hat{\beta}_1 (x_0 - \bar{x}).$$

This form is extremely revealing.

- Near $x_0 = \bar{x}$, the fitted value depends little on the slope because $(x_0 - \bar{x}) \approx 0$.
- Far from $\bar{x}$, small changes in $\hat{\beta}_1$ are multiplied by the large distance $(x_0 - \bar{x})$.

Therefore points in the tails are more sensitive to slope changes.

9.2 Variance explanation

The variance of the fitted mean is

$$\mathrm{Var}(\hat{y}(x_0)) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right).$$

As $|x_0 - \bar{x}|$ increases, the variance increases quadratically.

So tail predictions are less stable.

9.3 Geometric interpretation

The regression line is estimated to best fit the cloud of points. If most data lie in the middle, the line is "anchored" there. Near the center, many points constrain the fitted value. Near the extremes, there are fewer neighboring points, so the line can pivot more.

That pivoting is exactly why the fitted model is more sensitive in the tails.

10 Leverage and Pointwise Confidence Width

10.1 Definition of leverage

The diagonal elements of the hat matrix,

$$h_{ii} = H_{ii},$$

are called *leverages*.

In SLR with intercept,

$$\boxed{h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}}$$

This has the same form as the variance factor for the fitted mean at x_i.

10.2 Connection to the CI width

For a design point x_0, define its generalized leverage-like quantity

$$h(x_0) = x_0^\top (X^\top X)^{-1} x_0.$$

Then

$$SE_{\text{mean}}(x_0) = s\sqrt{h(x_0)},$$
$$SE_{\text{pred}}(x_0) = s\sqrt{1 + h(x_0)}.$$

So the pointwise confidence width is directly controlled by leverage.

- High leverage $\Rightarrow$ wider mean-response CI
- High leverage $\Rightarrow$ wider prediction interval

10.3 At observed data points

At an observed point x_i,

$$\text{Var}(\hat{y}_i) = \sigma^2 h_{ii}.$$

Thus high leverage points have more variable fitted values.

11 Standardized Residuals, Leverage, and Cook's Distance

11.1 Residual variance depends on leverage

The residual is

$$e_i = y_i - \hat{y}_i.$$

Its variance is

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii}).$$

Thus the standardized residual is

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}.$$

A large standardized residual means the point is unusual in the vertical direction after accounting for leverage.

11.2 Cook's distance

Cook's distance measures how much the fitted model changes if observation i is removed. A common formula is

$$\boxed{D_i = \frac{e_i^2}{ps^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}}$$

or equivalently in terms of standardized residuals,

$$\boxed{D_i = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}}$$

This shows that influence depends on both:

- vertical outlyingness: r_i^2
- horizontal extremeness: h_{ii}

11.3 Interpretation

- Large residual + low leverage: may be an outlier, but less influential on the line.
- Small residual + high leverage: may not look bad, but has potential to pull the line.
- Large residual + high leverage: most dangerous, often highly influential.

12 Why Tail Points Have High Influence Potential

In SLR,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}.$$

If x_i is far from $\bar{x}$, then h_{ii} is large.

Thus tail points tend to have high leverage. Since fitted means and interval widths depend on h_{ii}, these points strongly affect:

- slope estimation,
- fitted values near the tails,
- width of pointwise confidence bands,
- influence diagnostics such as Cook's distance.

12.1 Sensitivity of slope to a tail point

Recall

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}.$$

A point far from $\bar{x}$ contributes a large multiplier $(x_i - \bar{x})$. So if such a point has an unusual y_i, it can strongly change the numerator and hence the slope.

This is another direct proof that tail points can strongly affect the fitted line.

13 Numerical Example: Leverage and Influence

Suppose

$$n = 10, \quad \bar{x} = 5, \quad S_{xx} = 40, \quad s = 2, \quad p = 2.$$

Take two points:

$$x_A = 5 \quad \text{(center)}, \qquad x_B = 11 \quad \text{(tail)}.$$

Their leverages are:

$$h_A = \frac{1}{10} + \frac{(5-5)^2}{40} = 0.1,$$

$$h_B = \frac{1}{10} + \frac{(11-5)^2}{40} = 0.1 + \frac{36}{40} = 1.0.$$

This shows an extreme tail point can have very high leverage.

Now suppose both points have the same raw residual magnitude:

$$e_A = e_B = 2.$$

Standardized residuals:

$$r_A = \frac{2}{2\sqrt{1-0.1}} = \frac{2}{2\sqrt{0.9}} \approx 1.054,$$

but for h_B very near 1, the denominator is tiny, making the standardized residual very large and the point potentially highly influential.

Cook's distance also rises sharply because of the factor

$$\frac{h_{ii}}{(1-h_{ii})^2}.$$

14 R Simulation: CI for Coefficients and Pointwise Intervals

14.1 Simulation goals

We will simulate data from a linear model, fit it repeatedly, and examine:

- coverage of CI for β_1,
- width of mean-response CI and prediction interval at center and tail,
- how leverage increases near the tails.

14.2 R code

```
set.seed(123)

n <- 40
x <- seq(0, 20, length.out = n)
beta0 <- 5
beta1 <- 2
sigma <- 4

y <- beta0 + beta1 * x + rnorm(n, sd = sigma)
fit <- lm(y ~ x)

summary(fit)
confint(fit)

# Two x-values: center and tail
newdat <- data.frame(x = c(mean(x), max(x)))

# CI for expected response
pred_mean <- predict(fit, newdata = newdat, interval = "confidence")

# PI for new response
pred_obs  <- predict(fit, newdata = newdat, interval = "prediction")

pred_mean
pred_obs
```

```
# Leverages
h <- hatvalues(fit)
cbind(x = x, leverage = round(h, 4))

# Diagnostic measures
rstandard(fit)
cooks.distance(fit)

# Plot fitted line and intervals on a fine grid
grid <- data.frame(x = seq(min(x), max(x), length.out = 200))
ci_grid <- predict(fit, newdata = grid, interval = "confidence")
pi_grid <- predict(fit, newdata = grid, interval = "prediction")

plot(x, y, pch = 19, main = "Fitted␣line␣with␣CI␣and␣PI")
lines(grid$x, ci_grid[, "fit"], lwd = 2)
lines(grid$x, ci_grid[, "lwr"], lty = 2)
lines(grid$x, ci_grid[, "upr"], lty = 2)
lines(grid$x, pi_grid[, "lwr"], lty = 3)
lines(grid$x, pi_grid[, "upr"], lty = 3)
legend("topleft",
       legend = c("fit", "mean␣CI", "prediction␣PI"),
       lty = c(1,2,3), bty = "n")
```

14.3 What you should see

- The confidence interval for the mean is narrowest near $\bar{x}$.
- Both bands widen toward the tails.
- The prediction interval is always wider than the mean-response confidence interval.

15 R Simulation: Empirical Comparison of Widths at Center vs Tail

```
set.seed(1)

B <- 1000
n <- 30
beta0 <- 3
beta1 <- 1.5
sigma <- 3

x <- seq(0, 10, length.out = n)
x_center <- mean(x)
x_tail <- max(x)

width_mean_center <- numeric(B)
width_mean_tail   <- numeric(B)
width_pred_center <- numeric(B)
width_pred_tail   <- numeric(B)
```

```
for (b in 1:B) {
  y <- beta0 + beta1 * x + rnorm(n, sd = sigma)
  fit <- lm(y ~ x)

  pm_center <- predict(fit,
                       newdata = data.frame(x = x_center),
                       interval = "confidence")
  pm_tail   <- predict(fit,
                       newdata = data.frame(x = x_tail),
                       interval = "confidence")

  pp_center <- predict(fit,
                       newdata = data.frame(x = x_center),
                       interval = "prediction")
  pp_tail   <- predict(fit,
                       newdata = data.frame(x = x_tail),
                       interval = "prediction")

  width_mean_center[b] <- pm_center[,"upr"] - pm_center[,"lwr"]
  width_mean_tail[b]   <- pm_tail[,"upr"] - pm_tail[,"lwr"]
  width_pred_center[b] <- pp_center[,"upr"] - pp_center[,"lwr"]
  width_pred_tail[b]   <- pp_tail[,"upr"] - pp_tail[,"lwr"]
}

colMeans(cbind(width_mean_center,
               width_mean_tail,
               width_pred_center,
               width_pred_tail))
```

Expected pattern:

$$\text{mean CI at center} < \text{mean CI at tail} < \text{prediction PI at center} < \text{prediction PI at tail}$$

though the middle ordering may sometimes vary numerically, the prediction interval is always wider than the corresponding mean CI at the same x_0.

16 R Simulation: Sensitivity to a Tail Point

Now we illustrate how adding one extreme point can alter the fit.

```
set.seed(42)

n <- 30
x <- seq(0, 10, length.out = n)
y <- 2 + 1.5 * x + rnorm(n, sd = 1.5)

fit1 <- lm(y ~ x)

# Add one tail point
x2 <- c(x, 18)
y2 <- c(y, 40)   # unusual tail response
fit2 <- lm(y2 ~ x2)
```

```
par(mfrow = c(1,2))

plot(x, y, pch = 19, main = "Original fit")
abline(fit1, lwd = 2)

plot(x2, y2, pch = 19, main = "Fit after adding tail point")
abline(fit2, lwd = 2, col = 2)

par(mfrow = c(1,1))

summary(fit1)
summary(fit2)

hatvalues(fit2)
rstandard(fit2)
cooks.distance(fit2)
```

16.1 Interpretation

You will usually see that:

- the slope changes noticeably,
- the new extreme point has large leverage,
- if its residual is also large, Cook's distance becomes large,
- confidence bands near the tail become more affected.

17 Summary of Main Proofs and Results

1. CI for coefficients

$$\hat{\beta}_j \pm t_{\alpha/2,n-p}\, s\sqrt{c_{jj}}, \qquad c_{jj} = [(X^\top X)^{-1}]_{jj}.$$

2. Pointwise CI for expected response

$$\hat{\mu}(x_0) \pm t_{\alpha/2,n-p}\, s\sqrt{x_0^\top (X^\top X)^{-1} x_0}.$$

3. Pointwise prediction interval for new observation

$$\hat{\mu}(x_0) \pm t_{\alpha/2,n-p}\, s\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}.$$

4. Why the prediction interval is wider

Because of the extra variance term from the new observation:

$$\mathrm{Var}(Y_{\text{new}} - \hat{\mu}(x_0)) = \sigma^2 + \mathrm{Var}(\hat{\mu}(x_0)).$$

5. Why tail points matter more

In SLR,

$$\mathrm{Var}(\hat{y}(x_0)) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right),$$

which increases as $|x_0 - \bar{x}|$ grows.

6. Leverage

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$$

in SLR, so tail points have high leverage.

7. Standardized residuals and Cook's distance

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}, \qquad D_i = \frac{r_i^2}{p} \frac{h_{ii}}{1 - h_{ii}}.$$

Influence is largest when both leverage and residual size are large.

18 Pedagogical Conclusion

There are three related but distinct ideas:

1. **Uncertainty in coefficients**: measured by the CI for β_j.
2. **Uncertainty in the fitted mean**: measured by the pointwise CI for $E[Y \mid X = x_0]$.
3. **Uncertainty in an individual future response**: measured by the prediction interval for $Y \mid X = x_0$.

The further x_0 is from the center of the data, the larger the leverage-like term, and the less stable the fitted value becomes. This is why regression fits are most reliable near the center of the design and most fragile in the tails.